PRACTICAL-7

1. Aim

Interface RFID module with Raspberry Pi to read and write to RFID tags/cards

2. Prior Concepts

i) Concept of Python Programming.

ii) SPI Communication interface.

3. New Concepts

Interfacing RFID with Raspi.

4. Hardware Required

i) RFID RC522 Reader Module

ii)Few RFID Cards or RFID Tags

iii) Jumper Wires

5. Procedure

Pin Connection

RFID Module PIN	RPi Pin	Function
SDA	24	GPIO 8(SPIO CS0)
SCK	23	GPIO 11(SPIO SCLK)
MOSI	19	GPIO 10(SPIO MOSI)
MISO	21	GPIO 9(SPIO MISO)
GND	6	GND(Ground)
RST	22	GPIO 25
3.5v	1	3.3V

i) Install Python3 on Raspi.

Write the below code in command prompt:

sudo apt install idle3

Setting up Raspberry Pi for Serial Communication

sudo raspi-config

In the "Interfacing Options", select the "Serial" option.

Select "Yes" option.

Finish the process and Reboot the Raspberry Pi.

Following are the terminal command that needs to be executed.

sudo apt update

python3 -m pip install spidev

pip install mfrc522

6. Implementation with Python Code

Code:

```
import RPI.GPIO as GPIO
from mfrc522 import SimpleMFRC522
from time import sleep

reader = SimpleMFRC522()
print("Welcome to RFID Practical")
while True:
```

```
print("")
print ("Choose your Option")
print("1. Write your Tag")
print("2. Read your Tag")
print("3. Delete the content")
print("Press any key to Exit")
print("")
a=input("Enter your option:")

if a=="1":
        text=input("New data: ")
        print("Now Place your Tag to write")
        reader.write(text)
        print("Written")
        sleep(1)

elif a=="2":
        print("Now Place your Tag to read")
        id, text=reader.read()
        print(id)
        print(text)
        sleep(1)

elif a=="3":
        print("Now Place your Tag to Reset")
        reader.write(text)
        print("Data Deleted")
        sleep(1)
```

```
else:
        break

GPIO.cleanup()
```

[Note: now place the RFID cards one by one over RFID module to check the output

7. Results: